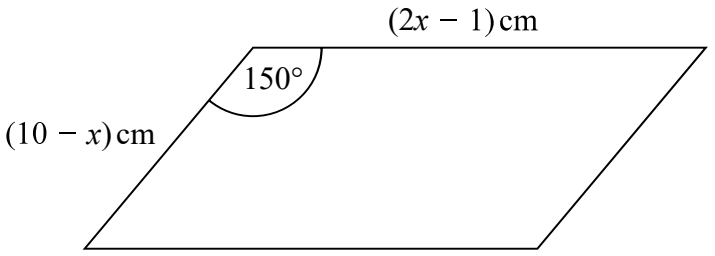
23 The diagram shows a parallelogram.



The area of the parallelogram is greater than 15 cm^2

(a) Show that $2x^2 - 21x + 40 < 0$

Area = (2x – 1)(10 – x)sin150°.

Since sin150° = 1/2, area = ½(2x – 1)(10 – x).

Area > 15, so (2x – 1)(10 – x) > 30.

Expand: $-2x^2 + 21x - 10 > 30$.

Rearrange: $2x^2 - 21x + 40 < 0$, as required.

3 marks total
Use area / sine
Expand accurately
Rearrange inequality

(3)

(b) Find the range of possible values of x.

Factor: (2x – 5)(x – 8) < 0.

Critical values are x = 2.5 and x = 8.

Product is negative between them: 2.5 < x < 8.

3 marks total
Factorise
Critical values
Between roots

.....

(3)

(Total for Question 23 is 6 marks)